RAMIPRIL

Cardiovascular protection

Teratogenic

CHEMICAL NAME (2S,3aS,6aS)-1-[(2S)-2-{[(2S)-1-ethoxy-1-oxo-4-phenylbutan-2-yl]amino}propanoyl]octahydrocyclopenta[b]pyrrole-2-carboxylic acid

CLASS ACE inhibitor (prodrug)

TRADE NAMES Tritace · Altace · Ramace · Triatec

THERAPEUTIC USES / INDICATIONS

- Hypertension.
- Prevention of cardiovascular events in high-risk patients (HOPE trial).
- Heart failure after myocardial infarction.
- Diabetic and non-diabetic nephropathy.

ADMINISTRATION

- **PO** capsules/tablets once daily (or twice daily) with or without food.
- Capsules may be opened and mixed with water or apple juice.

DOSAGES

- **Hypertension:** start 1.25–2.5 mg daily; usual 2.5–5 mg; max 10 mg/day.
- **CV prevention:** 2.5 mg daily, increase to 10 mg daily.
- **Post-MI heart failure:** 2.5 mg twice daily, titrate to 5 mg twice daily.
- CrCl <30: max 5 mg/day.

PHARMACOKINETICS

- **Onset:** 1–2 h. **Peak:** 3–6 h. **Duration:** 24 h.
- Prodrug converted in liver to ramiprilat.
- Excreted in urine. **t½:** 13–17 h (ramiprilat, terminal >50 h).

PHARMACODYNAMICS

- Inhibits **ACE** in plasma and tissues → ↓ angiotensin II and aldosterone.
- Improves endothelial function; reduces cardiovascular events.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry cough.
- Dizziness, headache, fatigue.
- Postural hypotension.

ADVERSE EFFECTS (SERIOUS)

- Angioedema.
- Hyperkalaemia.
- Acute renal failure.
- Neutropenia, pancreatitis (rare).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy and breastfeeding.
- Angioedema history; hereditary angioedema.
- Bilateral renal artery stenosis; haemodynamically unstable patients.
- Sacubitril/valsartan within 36 h.

KEY INTERACTIONS

- **K^+ supplements, spironolactone:** hyperkalaemia.
- **NSAIDs:** AKI, ↓ effect.
- **Lithium:** toxicity.
- **mTOR inhibitors, DPP-4 inhibitors:** ↑ angioedema.

NURSING MANAGEMENT

- Baseline and follow-up U&E.
- BP monitoring; postural BP.
- Assess for cough and angioedema.
- Hold if hypotension or hyperkalaemia.
- Educate on adherence (long-term protection).

PATIENT EDUCATION

- Take daily even if you feel well.
- Rise slowly.
- Report face/tongue swelling, fainting.
- Avoid pregnancy; avoid NSAIDs.

KEY POINTS

- Prodrug ACEi with proven CV protection.
- Cough, angioedema, hyperkalaemia.
- Target 10 mg daily in high-risk patients.
- Avoid in pregnancy.

SOURCES CHECKED **[273]** Ramipril 10mg Tablets (UK SmPC) **[274]** Ramipril, oral (US label) **[275]** Ramipril, oral (US label, Zydus Pharmaceuticals USA Inc)

Full references p. 432–457